

Zenghong He

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EDUCATION

University of California San Diego

B.S. Electrical Engineering – Machine Learning and Control

La Jolla, CA

Expected Dec 2026

RESEARCH INTERESTS

Robotic active matter and collective intelligence; nonlinear dynamics and emergent behavior in multi-agent systems; decentralized and learning-based control.

RESEARCH EXPERIENCE

UC San Diego — Research Assistant | PI: Prof. Tzer Han Tan

Nov 2024 – Present

Three Phases of Odd Robotic Active Matter

- Built and implemented custom MASBots and supporting experimental setups to study collective dynamics in nonreciprocal robotic active matter systems.
- Set up and calibrated experiments, and conducted systematic data recording for controlled multi-agent trials.
- Performed trajectory extraction, measurement, and data analysis to characterize MASBot interactions and collective behaviors.
- Modified and improved existing analysis code to process experimental data and examine dynamical patterns in the system.
- Contributed to data analysis and experimental validation for an ongoing research manuscript (see Publications).

Programmable Active Collectives with Long-Range Attraction Enable New States of Matter

(DARPA-funded research project)

- Participated in the design and implementation of slatted MASBots to study interaction effects in collective motion.
- Built and tested robotic systems, and conducted experiments to investigate collective transport and phase behaviors.
- Assisted in improving the tracking pipeline for experimental data acquisition and trajectory extraction.
- Contributed to experimental data processing and analysis to examine collective dynamics in the system.

PUBLICATIONS & MANUSCRIPTS

* Authors contributed equally † Corresponding authors

Fan Bo*, Shiqi Liu*, **Zenghong He**, Wyatt Joyce, Gregor Leech, Kiet Tran, Keilan Ramirez, Nicholas Boechler, Nicholas Gravish, Hongbo Zhao, Tzer Han Tan[†]. Three phases of odd robotic active matter. Submitted.

PROJECTS

Deep Convolutional GAN Implementation (PyTorch): Implemented and trained DCGAN models on CelebA and SUN datasets, evaluating generative performance using FID and exploring architectural and training improvements.

TECHNICAL SKILLS

Programming: C++, Python, MATLAB, Verilog, x86 Assembly

Modeling & Control: Linear control systems, optimization, nonlinear dynamics, data-driven modeling

Experimental & Hardware: Robotic prototyping, experimental system setup and calibration, motion tracking and trajectory analysis, circuit design, LTspice

Languages: English, Mandarin, Cantonese

TEACHING EXPERIENCE

De Anza College — Senior Tutor (Mathematics, Physics, Chemistry, Statistics) Apr 2023 – Jun 2024

- Provided individual and group instruction in quantitative subjects, supporting undergraduate students in problem-solving and conceptual understanding.
- Mentored and supervised junior tutors, coordinating instructional support across multiple STEM disciplines.